

Assignment12 – Deep Learning for NLP

Given: July 8 Due: July 13

Problem 12.1 (Continuous Bag of Words: Design Trade-Offs)

In the CBOW algorithm, explain the trade-offs regarding the

1. ...size of the context window
2. ...dimension of the hidden layer

Problem 12.2 (Back Propagation vs. Feedback)

We compare back propagation and feedback/recurrency in neural networks.

1. Explain (in about 1 sentence) their commonalities.
2. Explain (in about 2 sentences) their differences.

Problem 12.3 (Context Information in RNNs for NLP)

Using examples, explain the difficulty of taking context into account for NLP and how the following neural network design choices aim at tackling it.

1. Recurrency
2. Bidirectional RNNs
3. Long short-term memory